You need JavaScript enabled to use this service.

Cookies on Civil Service Jobs

We'd like to use analytics cookies so we can understand how you use Civil Service Jobs and make improvements.

We also use essential cookies to remember if you ve accepted analytics cookies.

5054f1e1bb39825d0a4a11628d52c393.1792

Accept analytics cookies

Reject analytics cookies

1792214722-3e992ca648937f5b81c9de2cd9

change
Language
121
BQsDAAAABgoFc2NvcmUAAAAKc2VhcmNoc
1224
yes
1792214722-3e992ca648937f5b81c9de2cd9

Genomics Bioinformatics Software Engineer

UK Health Security Agency

Apply before 11:55 pm on Friday 30th October 2026

UK Health Security Agency

Reference number

486142

Salary

£60,494 - £70,566

per annum, pro rata

A Civil Service Pension with an employer contribution of 28.97%

Job grade

Grade 7

Contract type

Fixed term
Loan
Secondment

Length of employment

to 31/03/2029

Type of role

Other

Working pattern

Flexible working, Full-time, Job share, Part-time

Number of jobs available

1

Contents

Location
About the job
Benefits
Things you need to know
Apply and further information

Location

London - Canary Wharf

About the job

Job summary

This is an analysis role within Genomics Public Health Analysis (GPHA), delivering cutting-edge analysis to support public health decision-making, with a focus on metagenomics, horizon scanning, emerging infections and developmental genomics.

The team works at pace to meet changing public health needs, including developing analysis and reporting for the Metagenomics Surveillance, Collaboration and Analysis Programme (mSCAPE), horizon scanning for emerging threats, developing analysis methods for pathogens, supporting national outbreaks and collaborating with epidemiology, modelling and academic partners.

We are seeking a highly motivated software developer with experience of genomic software solutions and bioinformatics pipelines. The priority is increasing the use of metagenomic data across UKHSA through the mSCAPE pilot, developing scalable solutions capable of processing high volumes of bacterial, viral and fungal sequence data.

Systems are primarily written in Python, with opportunities to use other languages. Software should be modular, documented and version-controlled, with workflow management or containerisation where appropriate, and deployable across different computational environments.

The role involves working closely with UKHSA, NHS and academic partners, with code shared through GitLab and GitHub.

Working for your organisation

We pride ourselves as being an employer of choice, where Everyone Matters promoting equality of opportunity to actively encourage applications from everyone, including groups currently underrepresented in our workforce.

UKHSA ethos is to be an inclusive organisation for all our staff and stakeholders. To create, nurture and sustain an inclusive culture, where differences drive innovative solutions to meet the needs of our workforce and

wider communities. We do this through celebrating and protecting differences by removing barriers and promoting equity and equality of opportunity for all.

Please visit our careers site for more information https://gov.uk/ukhsa/careers

Job description

- Contribute to the development and implementation of a strategy, jointly with partners, for creating software processes for metagenomic data processing, analysis, reporting and storage within mSCAPE
- To support delivery of routine metagenomic-based analysis including managing issues raised by stakeholders
- To coordinate the UKHSA contribution to development of new software and design with appropriate architecture for more efficient and scalable ways of automated processing, analysis, interpretation and visualisation of sequence data
- To work with bioinformatics colleagues to develop and deploy, as appropriate, analytical pipelines
- To ensure best practice for software development, testing and implementation is followed in mSCAPE
- To ensure compliance with quality procedures and achieve appropriately recognised software accreditation or compliance where relevant
- To support software improvements required to enhance current UKHSA service delivery

Please note this list is not exhaustive.

Person specification

Essential role criteria:

- Higher degree (i.e., MSc or above) in Computer Science, Bioinformatics, or related discipline or equivalent experience working in a Bioinformatics team.

Extensive experience leading the design, development and delivery of large and complex software solutions within secure, governed or regulated environments handling sensitive research, clinical or public health data

Demonstrable skills in developing APIs, web services and data-centric applications

Proven ability to write high-quality software documentation and implementation of quality standards for coding among a team including code review, unit testing and CI/CD principles.

Expert knowledge/experience of Unix, PostgreSQL or similar RDBMS. Proven expert skills in at least one object-oriented programming language, preferably Python

Knowledge of deploying code on distributed computational resources HPC, public / private cloud or TRE infrastructures and experience with containerisation technologies such as Docker and container orchestration platforms such as Kubernetes.

Proven skills in developing and modifying code used for data analysis

An understanding of and commitment to equality of opportunity and good working relationships, both in terms of day-to-day working practices, but also in relation to management systems

Desirable role criteria:

Proven experience acting as technical lead or principal engineer, influencing architecture, engineering standards and technical strategy across multi-disciplinary projects.

Experience of software used for pipeline development, e.g., NextFlow, Snakemake, Ansible

Experience using data processing and visualization packages such as pandas, matplotlib and Plotly

Experience of achieving software accreditation

Experience in industry or academia of delivering multi-disciplinary projects within deadlines

Experience of bioinformatics and an understanding of Next Generation Sequencing analytical techniques and advancements of medical microbiology and genomics.

Skills in project management of software delivery, in a collaborative environment using best practices concerning testing, documentation and version control using Git

An understanding of the main concepts of public health microbiology.

Benefits

Alongside your salary of £60,494, UK Health Security Agency contributes £17,525 towards you being a member of the Civil Service Defined Benefit Pension scheme. Find out what benefits a Civil Service Pension provides (opens in a new window).

Learning and development tailored to your role

An environment with flexible working options

A culture encouraging inclusion and diversity

A Civil Service pension with an employer contribution of 28.97%

Things you need to know

Artificial intelligence

Artificial intelligence can be a useful tool to support your application, however, all examples and statements provided must be truthful, factually accurate and taken directly from your own experience. Where plagiarism has been identified (presenting the ideas and experiences of others, or generated by artificial intelligence, as your own) applications may be withdrawn and internal candidates may be subject to disciplinary action. Please see our candidate guidance (opens in a new window) for more information on appropriate and inappropriate use.

Selection process details

This vacancy is using Success Profiles and will assess your Experience, Behaviours, and Technical skills.

Stage 1: Application & Sift

Success Profiles - GOV.UK

You will be required to complete an application form. You will be assessed on the listed 8 essential criteria, and this will be in the form of a:

Application form (Employer/ Activity history section on the application)

750 word supporting statement

This should outline how you consider your skills, experience and knowledge provide evidence of your suitability for the role, with reference to the essential criteria.

You will receive a joint score for your application form and statement. (Please note you will not be able to upload your CV. You must complete the application form in as much detail as possible. Please do not email us your CV).

Healthjobs UK has a word limit of 1500, but your supporting statement must be no more than 750 words. Please do not exceed 750 words.

Longlisting: In the event of a large number of applications we will longlist into 3 piles of:

1. Meets all essential criteria
2. Meets some essential criteria
3. Meets no essential criteria

If used, the pile Meets all essential criteria will proceed to shortlisting.

Shortlisting: In the event of a large number of applications we will conduct an initial sift, on the lead criteria of:

Demonstrable skills in developing APIs, web services and data-centric applications

Proven ability to write high-quality software documentation and implementation of quality standards for coding among a team including code review, unit testing and CI/CD principles.

Expert knowledge/experience of Unix, PostgreSQL or similar RDBMS. Proven expert skills in at least one object-oriented programming language, preferably Python

Desirable criteria may be used in the event of a large number of applications/ large amount of successful candidates.

If you are successful at this stage, you will progress to interview & assessment.

Feedback will not be provided at this stage.

Stage 2: Interview

Success Profiles - GOV.UK

You will be invited to a remote interview. Behaviours and Technical skills will be tested at interview. The Behaviours tested during the interview stage will be:

- Managing a Quality Service (Lead behaviour)
- Changing and Improving
- Seeing the Bigger Picture
- Communicating and influencing

Technical skills will be assessed through a presentation. There will be a 10-minute presentation on the topic of Enhancing a Genomics API to Support Advanced Analysis Queries.

Interview dates to be confirmed.

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Additional Information

Location

This role is being offered as hybrid working based at our core HQ Canary Wharf. We offer great flexible working opportunities at UKHSA and operate using a hybrid working model where business needs allow. This provides us with greater flexibility about how and where we work, to get the best from our workforce. As a hybrid worker, you will be expected to spend a minimum of 60% of your contractual working hours (approximately 3 days a week pro

rata, (averaged over a month) working at our office. Our core HQ offices are modern and newly refurbished with excellent city centre transport links and benefit from co-location with other government departments such as the Department for Health and Social Care (DHSC).

Security Clearance Level Requirement

Successful candidates must pass a basic disclosure and barring security check before they can be appointed.

Successful candidates must meet the security requirements before they can be appointed. The level of security needed is Baseline Personnel Security Standard.

Eligibility Criteria

Open to all external applicants (anyone) from outside the Civil Service (including internal applicants).

Salary Information

If you are successful at interview and are moving from another government department, NHS, or Local Authority, the relevant starting salary principles for level transfers or promotions will apply. Otherwise, roles are offered at the pay scale minimum for the grade, but in exceptional circumstances there may be flexibility if you are able to demonstrate you are already in receipt of an existing, higher salary. Pay increases are through the relevant annual pay award for the role and terms.

Reserve List

If more than the required number of suitable candidates pass the interview criteria, you may be kept on a reserve list for 12 months subject to your agreement. You may be contacted, in merit-order, if similar roles with closely matching essential criteria become available and the department choose to appoint from a reserve list.

The panel will assess if candidates meet the requirement of the role first, using a specific benchmark system. If you are interviewed for the post and do not meet the required threshold for the specified grade, your application may

be assessed against a similar, lower grade role and you may be offered the post should one be available.

Incident Response Activity

From time to time, the role holder may be required to support UKHSA s incident response activity by contributing their appropriate expertise within an Incident Cell; during such periods, supported by the line manager, this work will take priority over normal day-to-day responsibilities, commensurate with their grade, skills and experience.

Qualifications and Registrations

For roles where specific qualifications or registrations are required, successful applicants will be asked to provide appropriate evidence. Employment cannot commence until satisfactory documentation has been received and verified.

Future location

UKHSA is investing in a new state-of-the-art National Biosecurity Centre in Harlow, Essex, which will eventually bring together teams currently based at Canary Wharf, Colindale and Porton Down. For more details, please see: Huge biosecurity centre investment to boost pandemic protection - GOV.UK. The new facilities will start becoming operational in the mid-2030s, with full completion by 2038. Staff will move in phases as facilities become available. If you're appointed to a role currently based at Canary Wharf, Colindale or Porton Down, please note that we'll continue investing in these sites for the next decade. As we get closer to the transition, we'll provide full information about relocation support available to staff.

Reasonable Adjustments

The Civil Service is committed to making sure that our selection methods are fair to everyone. To help you during the recruitment process, we will consider any reasonable adjustments that could help you. An adjustment is a change to the recruitment process or an adjustment at work. This is separate to the Disability Confident Scheme. If you need an adjustment to be made at any point during the recruitment process you should contact the recruitment team in confidence as soon as possible to discuss your needs.

You can find out more information about reasonable adjustments across the Civil Service here: https://www.civil-service-careers.gov.uk/reasonable-adjustments/

International Police check

If you have spent more than 6 months abroad over the last 3 years you may need an International Police Check. This would not necessarily have to be in a single block, and it could be time accrued over that period.

Internal Fraud Check

If successful for this role as one aspect of pre-employment screening, applicant s personal details name, national insurance number and date of birth - will be checked against the Cabinet Office Internal Fraud Hub, and anyone included on the database will be refused employment unless they can show exceptional circumstances. Currently this is only for external candidates to the Civil Service.

If this role is fixed term and you are not currently a civil servant, you will take up the post on a Fixed Term appointment; you may be able to take this role up as a Secondment. If you are an existing Civil Servant, based outside of UKHSA, you will take up the post as a loan which you will need your department to agree. Loans are normally agreed for a period of two years. However, this may be extended where required, subject to agreement between UKHSA and your home department. You cannot take the post up as a fixed term. If you are an existing UKHSA member of staff, you will take up the post as either a level transfer or a temporary promotion as per the UKHSA s Pay policy. **If this role is permanent**, it may also be available as a loan (for existing civil servants only) or secondment arrangement, subject to business need and agreement from the hiring manager and, where applicable, the employee's home organisation.

Feedback will only be provided if you attend an interview or assessment.

Security

Successful candidates must undergo a basic (or equivalent) criminal record check.

People working with government assets must complete baseline personnel security standard (opens in new window) checks.

Nationality requirements

This job is broadly open to the following groups:

- UK nationals
- nationals of the Republic of Ireland
- nationals of Commonwealth countries who have the right to work in the UK
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities with settled or pre-settled status under the European Union Settlement Scheme (EUSS) (opens in a new window)
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities who have made a valid application for settled or pre-settled status under the European Union Settlement Scheme (EUSS)
- individuals with limited leave to remain or indefinite leave to remain who were eligible to apply for EUSS on or before 31 December 2020
- Turkish nationals, and certain family members of Turkish nationals, who have accrued the right to work in the Civil Service

Further information on nationality requirements (opens in a new window)

Working for the Civil Service

The Civil Service Code (opens in a new window) sets out the standards of behaviour expected of civil servants.

We recruit by merit on the basis of fair and open competition, as outlined in the Civil Service Commission's recruitment principles (opens in a new window).

The Civil Service embraces diversity and promotes equal opportunities. As such, we run a Disability Confident Scheme (DCS) for candidates with disabilities who meet the minimum selection criteria.

Diversity and Inclusion

The Civil Service is committed to attract, retain and invest in talent wherever it is found. To learn more please see the Civil Service People Plan (opens in a new window) and the Civil Service Diversity and Inclusion Strategy (opens in a new window).

Apply and further information

This vacancy is part of the Great Place to Work for Veterans (opens in a new window) initiative.

The Civil Service welcomes applications from people who have recently left prison or have an unspent conviction. Read more about prison leaver recruitment (opens in new window).

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Contact point for applicants

Job contact :

Name : Tom Geldeard

Email : tom.geldeard@ukhsa.gov.uk

Recruitment team

Email : recruitment@ukhsa.gov.uk

Further information

The law requires that selection for appointment to the Civil Service is on merit on the basis of fair and open competition as outlined in the Civil Service Commission's Recruitment Principles. If you feel your application has not been treated in accordance with the recruitment Principles, and you wish to make a complaint, in the first instance, you should contact the UKHSA Complaints team via email at: Complaints@ukhsa.gov.uk

If you are not satisfied with the response you receive from the Department, you can contact the Civil Service Commission. Visit the Civil Service Commission website for further information: https://

civilservicecommission.independent.gov.uk

Civil Service Commission

